Questions

Q1.

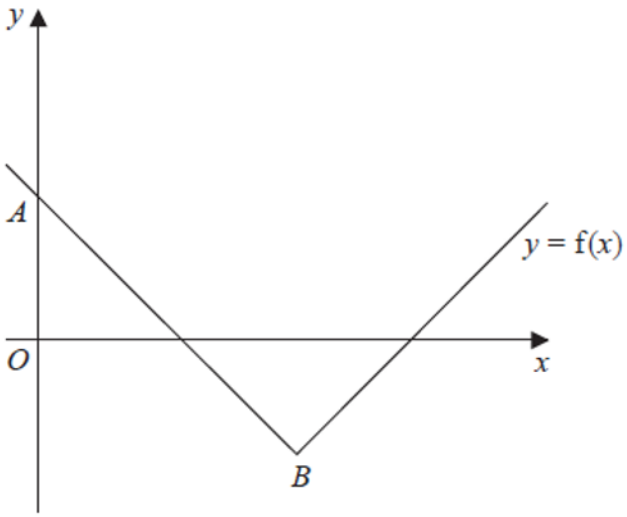


Figure 2

Figure 2 shows part of the graph with equation $y = f(x)$, where

$$f(x) = \tfrac{1}{2}kx - 9\tfrac{1}{2} - 2 \qquad x \in \mathbb{R}$$

and k is a positive constant.

The graph intersects the y-axis at the point A and has a minimum point at B as shown.

(a) (i) Find the y coordinate of A

(ii) Find, in terms of k, the x coordinate of B

(2)

(b) Find, in terms of k, the range of values of x that satisfy the inequality

$$|kx - 9| - 2 < 0$$

(3)

Given that the line $y = 3 - 2x$ intersects the graph $y = f(x)$ at two distinct points,

(c) find the range of possible values of k

(3)

(Total for question = 8 marks)

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(**Q05 WMA13/01, June 2022**)

Q2.

The point $P(-4, -3)$ lies on the curve with equation $y = f(x)$, $x \in \mathbb{R}$

Find the point to which P is mapped when the curve with equation $y = f(x)$ is transformed to the curve with equation

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(a) $y = f(2x)$

(1)

(b) $y = 3f(x - 1)$

(2)

(c) $y = |f(x)|$

(1)

(Total for question = 4 marks)

(**Q01 WMA13/01, Jan 2024**)

Q3.

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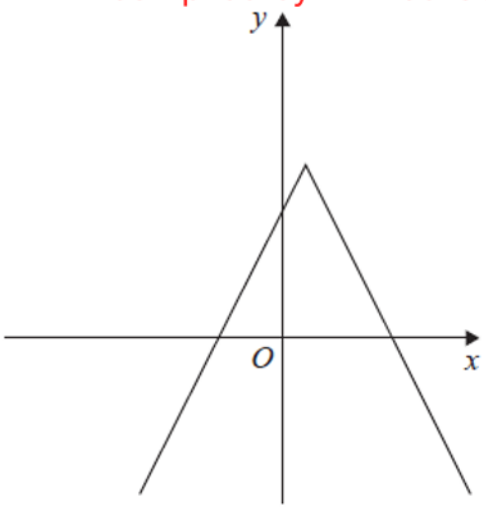


Figure 2

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In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The graph shown in Figure 2 has equation

$$y = a - |2x - b|$$

where a and b are positive constants, $a > b$

(a) Find, giving your answer in terms of a and b,

(i) the coordinates of the maximum point of the graph,

(ii) the coordinates of the point of intersection of the graph with the y-axis,

(iii) the coordinates of the points of intersection of the graph with the x-axis.

(5)

(b) On Diagram 1, sketch the graph with equation

$$y = |x| - 1$$

(2)

Given that the graphs $y = |x| - 1$ and $y = a - |2x - b|$ intersect at $x = -3$ and $x = 5$

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(c) find the value of a and the value of b

(4)

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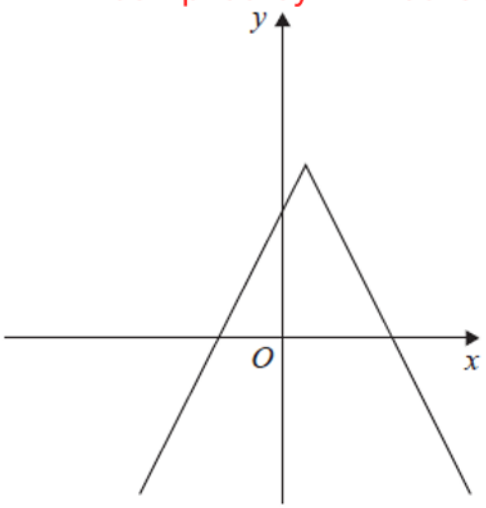


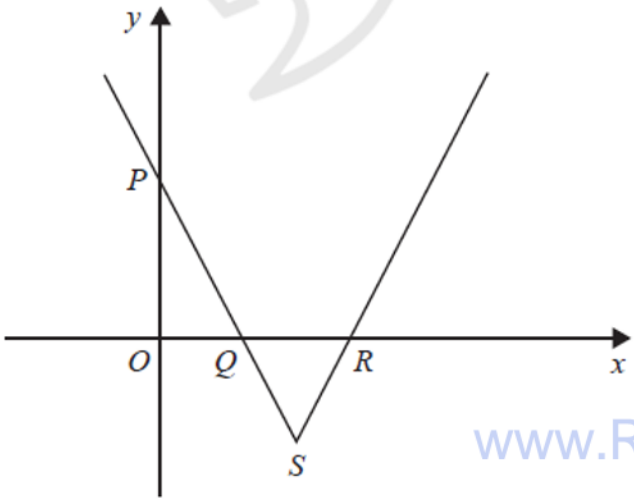
Diagram 1

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(Total for question = 11 marks)

(Q08 WMA13/01, Jan 2024)

Q4.



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Figure 2

Figure 2 shows a sketch of the graph with equation

$$y = |3x - 5a| - 2a$$

where a is a positive constant.

The graph

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- cuts the y-axis at the point P
- cuts the x-axis at the points Q and R
- has a minimum point at S

(a) Find, in simplest form in terms of a, the coordinates of

(i) point P

(ii) points Q and R

(iii) point S

(4)

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(b) Find, in simplest form in terms of a, the values of x for which

$$|3x - 5a| - 2a = |x - 2a|$$

(4)

(Total for question = 8 marks)

(**Q06 WMA13/01, Jan 2023**)

Q5.

The function f is defined by

$$f(x) = 2x^2 - 5 \qquad x \geqslant 0 \quad x \in \mathbb{R}$$

(a) State the range of f

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(1)

Below there is a diagram, labelled Diagram 1, which shows a sketch of the curve with equation $y = f(x)$.

(b) On Diagram 1, sketch the curve with equation $y = f^{-1}(x)$.

(2)

The curve with equation $y = f(x)$ meets the curve with equation $y = f^{-1}(x)$ at the point P

Using algebra and showing your working,

(c) find the exact x coordinate of P

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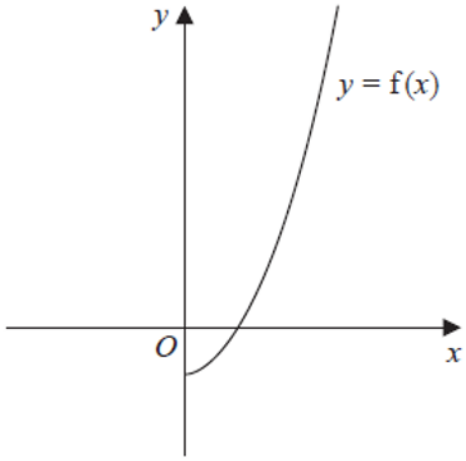


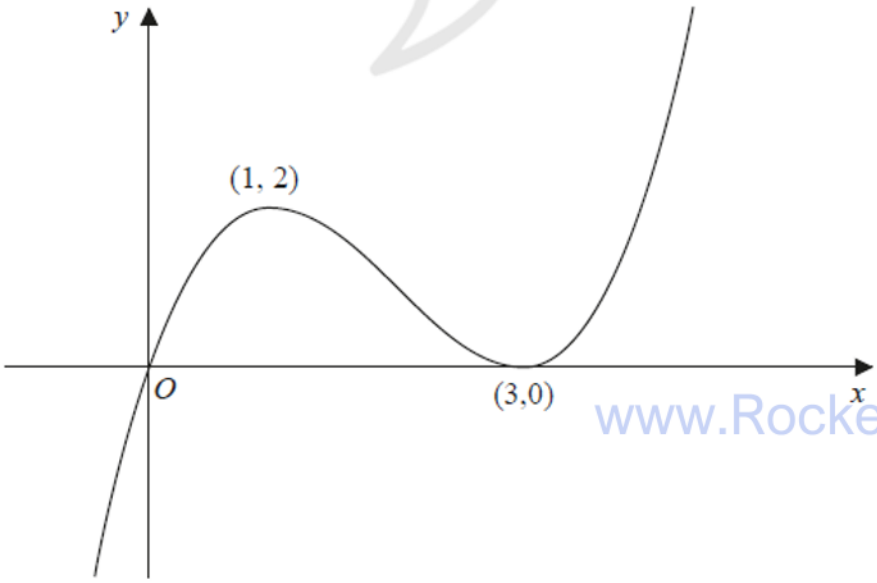
Diagram 1

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(Total for question = 6 marks)

(**Q04 WMA13/01, June 2023**)

Q6.



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Figure 1

Figure 1 shows a sketch of the curve with equation $y = f(x)$, where $x \in \mathbb{R}$ and $f(x)$ is a polynomial.

The curve passes through the origin and touches the x-axis at the point (3, 0)

There is a maximum turning point at (1, 2) and a minimum turning point at (3, 0)

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On separate diagrams, sketch the curve with equation

(i) $y = 3f(2x)$

(3)

(ii) $y = f(-x) - 1$

(3)

On each sketch, show clearly the coordinates of

- the point where the curve crosses the y-axis
- any maximum or minimum turning points

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(Total for question = 6 marks)

(**Q02 WMA13/01, Jan 2021**)

Q7.

The functions f and g are defined by

$$f(x) = \frac{4x + 6}{x - 5} \qquad x \in \mathbb{R},\ x \neq 5$$

$$g(x) = 5 - 2x^2 \qquad x \in \mathbb{R},\ x \leqslant 0$$

(a) Solve the equation

$$fg(x) = 3$$

(4)

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(b) Find f^{-1}

(3)

(c) Sketch and label, on the same axes, the curve with equation $y = g(x)$ and the curve with equation $y = g^{-1}(x)$. Show on your sketch the coordinates of the points where each curve meets or cuts the coordinate axes.

(3)

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(Total for question = 10 marks)

(Q04 WMA13/01, June 2021)

Q8.

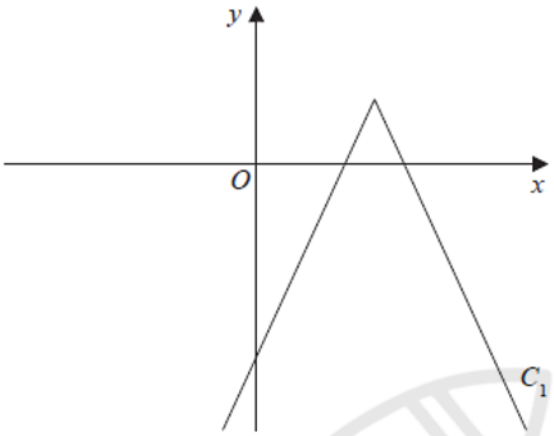


Figure 3

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Figure 3 shows a sketch of the graph of C_1 with equation

$$y = 5 - |3x - 22|$$

(a) Write down the coordinates of

(i) the vertex of C_1

(ii) the intersection of C_1 with the y-axis.

(2)

(b) Find the x coordinates of the intersections of C_1 with the x-axis.

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(2)

Diagram 1 is a copy of Figure 3.

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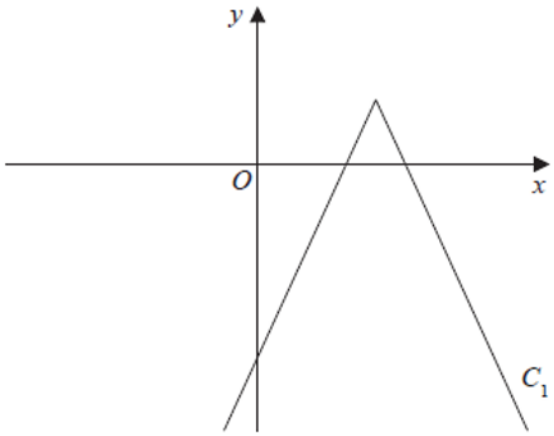


Diagram 1

(c) On Diagram 1, sketch the curve C_2 with equation

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$$y = \frac{1}{9}x^2 - 9$$

Identify clearly the coordinates of any points of intersection of C_2 with the coordinate axes.

(3)

(d) Find the coordinates of the points of intersection of C_1 and C_2

(*Solutions relying entirely on calculator technology are not acceptable.*)

(5)

(Total for question = 12 marks)

(**Q07 WMA13/01, Oct 2022**)

Q9.

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In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

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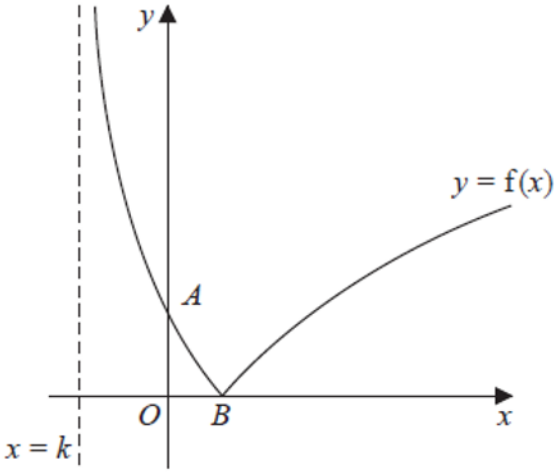


Figure 2

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Figure 2 shows a sketch of the curve with equation

$$y = |2 - 4\ln(x + 1)| \qquad x > k$$

where k is a constant.

Given that the curve

- has an asymptote at $x = k$
- cuts the y-axis at point A
- meets the x-axis at point B

as shown in Figure 2,

(a) state the value of k

(1)

(b) (i) find the y coordinate of A

(ii) find the exact x coordinate of B

(3)

(c) Using algebra and showing your working, find the set of values of x such that

$$|2 - 4\ln(x + 1)| > 3$$

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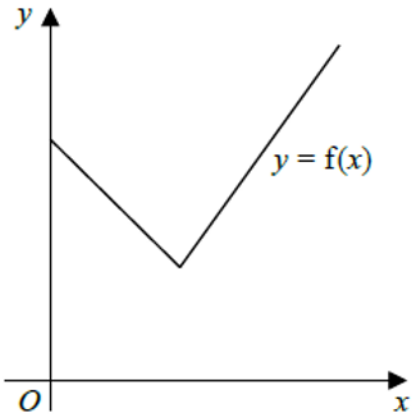
(5)

(Total for question = 9 marks)

(Q09 WMA13/01, Oct 2023)

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Q10.



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Figure 1

Figure 1 shows a sketch of part of the graph $y = f(x)$ where

$$f(x) = 2|3 - x| + 5 \qquad x \geqslant 0$$

(a) Solve the equation

$$f(x) = \frac{1}{2}x + 30$$

(3)

Given that the equation $f(x) = k$, where k is a constant, has two distinct roots,

(b) state the set of possible values for k.

(2)

(Total for question = 5 marks)

(Q03 WMA13/01, Specimen papers)

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Q11.

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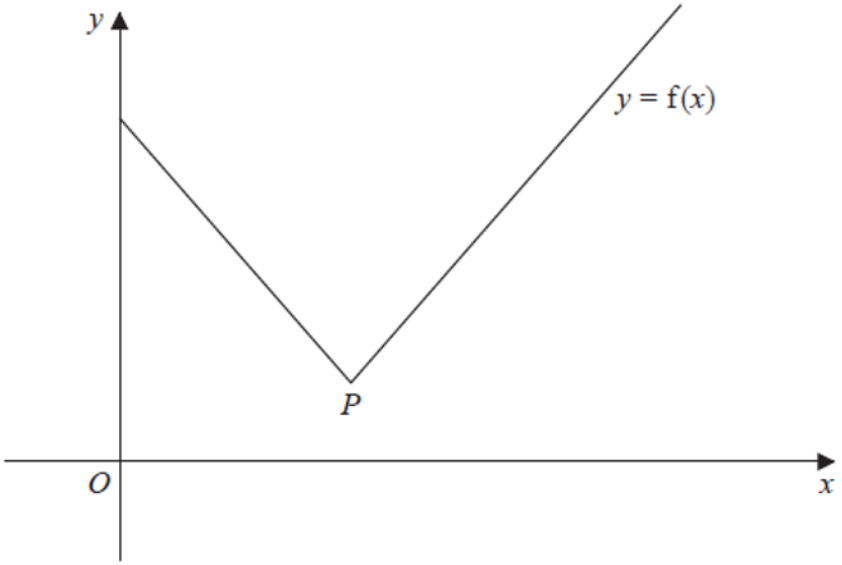


Figure 2

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Figure 2 shows part of the graph with equation $y = f(x)$, where

$$f(x) = 2|2x - 5| + 3 \qquad x \geqslant 0$$

The vertex of the graph is at point P as shown.

(a) State the coordinates of P.

(2)

(b) Solve the equation $f(x) = 3x - 2$

(4)

Given that the equation

$$f(x) = kx + 2$$

where k is a constant, has exactly two roots,

(c) find the range of values of k.

(3)

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(Total for question = 9 marks)

(Q06 WMA13/01, Jan 2020)

Q12.

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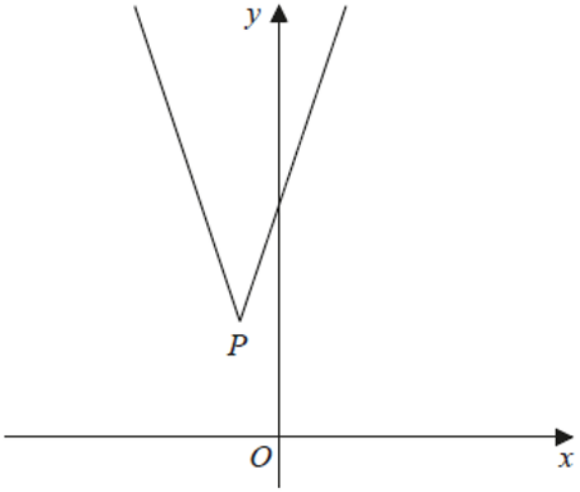


Figure 2

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Figure 2 shows a sketch of the graph with equation $y = f(x)$, where

$$f(x) = | 3x + a | + a$$

and where a is a positive constant.

The graph has a vertex at the point P, as shown in Figure 2.

(a) Find, in terms of a, the coordinates of P.

(2)

(b) Sketch the graph with equation $y = g(x)$, where

$$g(x) = | x + 5a |$$

On your sketch, show the coordinates, in terms of a, of each point where the graph cuts or meets the coordinate axes.

(2)

The graph with equation $y = g(x)$ intersects the graph with equation $y = f(x)$ at two points.

(c) Find, in terms of a, the coordinates of the two points.

(5)

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(Total for question = 9 marks)

(Q04 WMA13/01, Jan 2021)

Q13.

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Given that k is a positive constant,

(a) on separate diagrams, sketch the graph with equation

(i) $y = k - 2|x|$

(ii) $y = \left|2x - \frac{k}{3}\right|$

Show on each sketch the coordinates, in terms of k, of each point where the graph meets or cuts the axes.

(4)

(b) Hence find, in terms of k, the values of x for which

$$\left|2x - \frac{k}{3}\right| = k - 2|x|$$

giving your answers in simplest form.

(4)

(Total for question = 8 marks)

(Q06 WMA13/01, June 2021)

Q14.

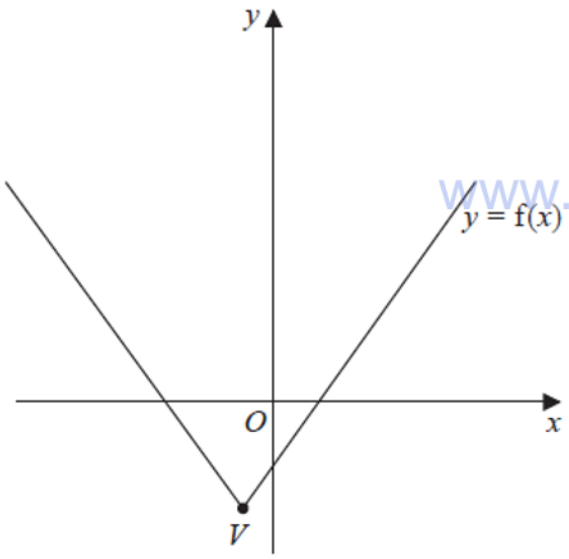


Figure 2

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Figure 2 shows a sketch of part of the graph with equation $y = f(x)$, where

$$f(x) = \frac{1}{2}|2x + 7| - 10$$

(a) State the coordinates of the vertex, V, of the graph.

(2)

(b) Solve, using algebra,

$$\frac{1}{2}|2x + 7| - 10 \geqslant \frac{1}{3}x + 1$$

(4)

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(c) Sketch the graph with equation

$$y = |f(x)|$$

stating the coordinates of the local maximum point and each local minimum point.

(4)

(Total for question = 10 marks)

(Q07 WMA13/01, Jan 2022)

Q15.

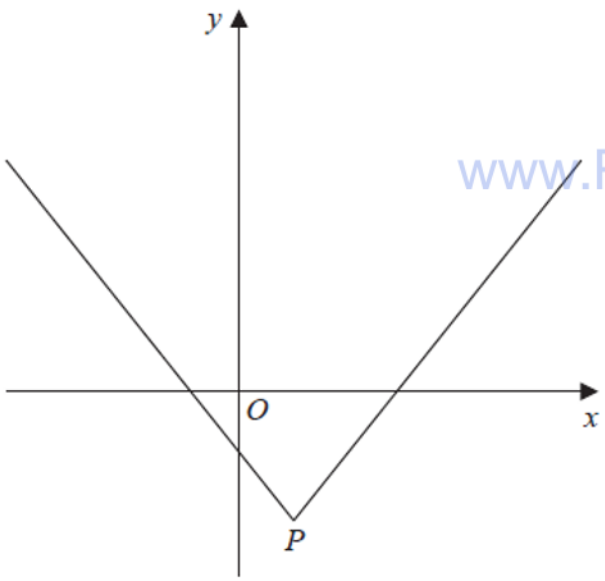


Figure 2

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Figure 2 shows a sketch of the graph $y = f(x)$, where

$$f(x) = 3|x - 2| - 10$$

The vertex of the graph is at point P, shown in Figure 2.

(a) Find the coordinates of P

(2)

(b) Find ff(0)

(2)

(c) Solve the inequality

$$3|x - 2| - 10 < 5x + 10$$

(2)

(d) Solve the equation

$$f(|x|) = 0$$

(3)

(Total for question = 9 marks)

(**Q06 WMA13/01, June 2023**)

Q16.

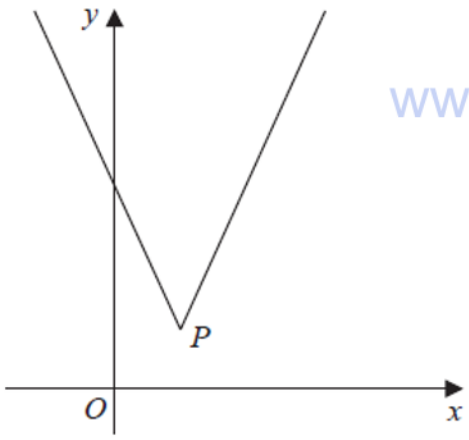


Figure 1

Figure 1 shows a sketch of part of the graph with equation $y = f(x)$, where

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$$f(x) = |3x - 13| + 5 \qquad x \in \mathbb{R}$$

The vertex of the graph is at point P, as shown in Figure 1.

(a) State the coordinates of P.

(2)

(b) (i) State the range of f.

(ii) Find the value of ff(4)

(2)

(c) Solve, using algebra and showing your working,

$$16 - 2x > |3x - 13| + 5$$

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(4)

The graph with equation $y = f(x)$ is transformed onto the graph with equation $y = af(x + b)$

The vertex of the graph with equation $y = af(x + b)$ is (4, 20)

Given that a and b are constants,

(d) find the value of a and the value of b.

(2)

(Total for question = 10 marks)

(Q02 WMA13/01, Oct 2021)

Q17.

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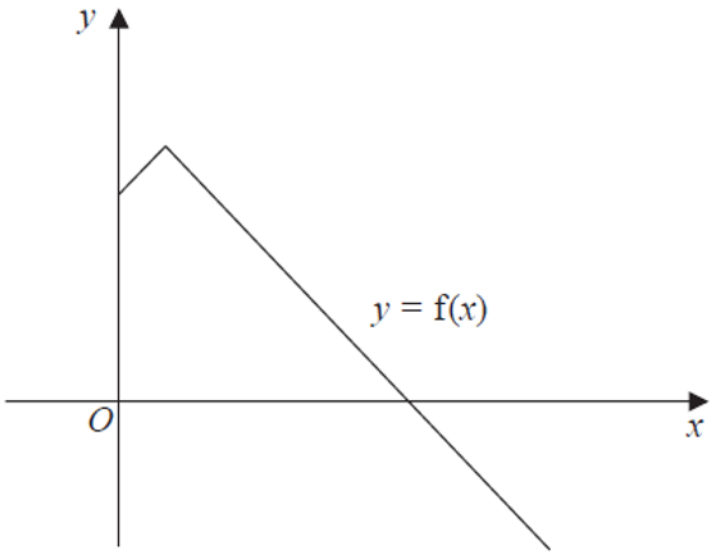


Figure 2

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Figure 2 shows a sketch of part of the graph with equation y = f(x) where

$$f(x) = 21 - 2|2 - x| \qquad x \geqslant 0$$

(a) Find ff(6)

(2)

(b) Solve the equation f(x) = 5x

(2)

Given that the equation f(x) = k , where k is a constant, has exactly two roots,

(c) state the set of possible values of k.

(2)

The graph with equation y = f(x) is transformed onto the graph with equation $y = a\,f(x - b)$

The vertex of the graph with equation $y = a\,f(x - b)$ is (6, 3).

Given that a and b are constants,

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(d) find the value of a and the value of b.

(2)

(Total for question = 8 marks)

(Q04 WMA13/01, Oct 2020)

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